

# VESSEL ANALYSIS REPORT

| vaf(%) | branch points count | tip point count | vein count | total vein length | average vein length |
|--------|---------------------|-----------------|------------|-------------------|---------------------|
| 2.135  | 88.0                | 171.0           | 217.0      | 6199.113          | 28.567              |

| id | p1.x, p1.y | p2.x, p2.y | length  | p1_type | p2_type |
|----|------------|------------|---------|---------|---------|
| 1  | [282,4]    | [277,6]    | 4.828   | TIP     | BRANCH  |
| 2  | [270,1]    | [277,6]    | 8.071   | TIP     | BRANCH  |
| 3  | [277,6]    | [277,27]   | 20.0    | BRANCH  | BRANCH  |
| 4  | [227,4]    | [215,10]   | 13.485  | TIP     | BRANCH  |
| 5  | [215,10]   | [109,71]   | 163.397 | BRANCH  | BRANCH  |
| 6  | [206,4]    | [215,10]   | 10.071  | TIP     | BRANCH  |
| 7  | [27,15]    | [24,66]    | 69.385  | BRANCH  | BRANCH  |
| 8  | [27,15]    | [53,47]    | 42.941  | BRANCH  | BRANCH  |
| 9  | [27,15]    | [56,21]    | 31.728  | BRANCH  | BRANCH  |
| 10 | [174,49]   | [146,17]   | 73.627  | TIP     | BRANCH  |
| 11 | [123,40]   | [146,17]   | 35.042  | TIP     | BRANCH  |
| 12 | [124,0]    | [146,17]   | 29.971  | TIP     | BRANCH  |
| 13 | [78,7]     | [77,19]    | 12.243  | TIP     | BRANCH  |
| 14 | [77,19]    | [56,21]    | 20.414  | BRANCH  | BRANCH  |
| 15 | [77,19]    | [83,34]    | 16.071  | BRANCH  | BRANCH  |
| 16 | [56,21]    | [53,47]    | 27.071  | BRANCH  | BRANCH  |
| 17 | [294,41]   | [277,27]   | 24.142  | TIP     | BRANCH  |
| 18 | [277,27]   | [257,80]   | 60.284  | BRANCH  | BRANCH  |
| 19 | [446,1]    | [457,27]   | 29.556  | TIP     | BRANCH  |
| 20 | [472,30]   | [457,27]   | 15.243  | TIP     | BRANCH  |
| 21 | [423,36]   | [457,27]   | 38.142  | TIP     | BRANCH  |
| 22 | [83,34]    | [104,63]   | 44.314  | BRANCH  | BRANCH  |
| 23 | [84,46]    | [83,34]    | 11.828  | TIP     | BRANCH  |
| 24 | [53,47]    | [37,66]    | 25.799  | BRANCH  | BRANCH  |
| 25 | [510,17]   | [501,50]   | 39.971  | TIP     | BRANCH  |
| 26 | [505,49]   | [501,50]   | 3.414   | TIP     | BRANCH  |

| id | p1.x, p1.y | p2.x, p2.y | length  | p1_type | p2_type |
|----|------------|------------|---------|---------|---------|
| 27 | [501,50]   | [471,83]   | 47.355  | BRANCH  | BRANCH  |
| 28 | [104,63]   | [109,71]   | 9.657   | BRANCH  | BRANCH  |
| 29 | [104,63]   | [71,99]    | 58.213  | BRANCH  | BRANCH  |
| 30 | [24,66]    | [37,66]    | 12.0    | BRANCH  | BRANCH  |
| 31 | [24,66]    | [21,115]   | 60.355  | BRANCH  | BRANCH  |
| 32 | [37,66]    | [49,94]    | 31.971  | BRANCH  | BRANCH  |
| 33 | [94,250]   | [109,71]   | 206.095 | TIP     | BRANCH  |
| 34 | [252,81]   | [257,80]   | 4.414   | TIP     | BRANCH  |
| 35 | [259,92]   | [257,80]   | 11.414  | TIP     | BRANCH  |
| 36 | [402,59]   | [471,83]   | 90.77   | TIP     | BRANCH  |
| 37 | [471,83]   | [481,98]   | 17.728  | BRANCH  | BRANCH  |
| 38 | [323,3]    | [311,92]   | 95.456  | TIP     | BRANCH  |
| 39 | [307,93]   | [311,92]   | 3.414   | TIP     | BRANCH  |
| 40 | [313,93]   | [311,92]   | 1.0     | TIP     | BRANCH  |
| 41 | [49,94]    | [71,99]    | 23.071  | BRANCH  | BRANCH  |
| 42 | [49,94]    | [21,115]   | 35.284  | BRANCH  | BRANCH  |
| 43 | [481,98]   | [499,125]  | 38.142  | BRANCH  | BRANCH  |
| 44 | [481,98]   | [486,124]  | 27.485  | BRANCH  | BRANCH  |
| 45 | [71,99]    | [76,144]   | 59.284  | BRANCH  | BRANCH  |
| 46 | [21,115]   | [30,145]   | 32.314  | BRANCH  | BRANCH  |
| 47 | [486,124]  | [499,125]  | 12.414  | BRANCH  | BRANCH  |
| 48 | [423,141]  | [486,124]  | 70.284  | TIP     | BRANCH  |
| 49 | [510,151]  | [499,125]  | 30.142  | TIP     | BRANCH  |
| 50 | [256,108]  | [256,135]  | 27.657  | TIP     | BRANCH  |
| 51 | [242,140]  | [256,135]  | 14.657  | TIP     | BRANCH  |
| 52 | [252,165]  | [256,135]  | 32.728  | TIP     | BRANCH  |
| 53 | [63,117]   | [76,144]   | 31.971  | TIP     | BRANCH  |
| 54 | [76,144]   | [74,162]   | 17.828  | BRANCH  | BRANCH  |
| 55 | [19,156]   | [30,145]   | 15.142  | TIP     | BRANCH  |
| 56 | [30,145]   | [43,161]   | 21.142  | BRANCH  | BRANCH  |
| 57 | [43,161]   | [74,162]   | 30.414  | BRANCH  | BRANCH  |
| 58 | [40,166]   | [43,161]   | 5.243   | TIP     | BRANCH  |

| id | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|----|------------|------------|--------|---------|---------|
| 59 | [74,162]   | [95,180]   | 27.042 | BRANCH  | BRANCH  |
| 60 | [472,161]  | [477,162]  | 4.414  | TIP     | BRANCH  |
| 61 | [477,162]  | [479,183]  | 20.828 | BRANCH  | BRANCH  |
| 62 | [486,155]  | [477,162]  | 10.485 | TIP     | BRANCH  |
| 63 | [96,197]   | [95,180]   | 16.414 | TIP     | BRANCH  |
| 64 | [107,168]  | [95,180]   | 18.485 | TIP     | BRANCH  |
| 65 | [479,183]  | [423,227]  | 80.426 | BRANCH  | BRANCH  |
| 66 | [479,183]  | [491,228]  | 59.184 | BRANCH  | BRANCH  |
| 67 | [70,213]   | [24,214]   | 54.385 | BRANCH  | BRANCH  |
| 68 | [68,209]   | [70,213]   | 3.414  | TIP     | BRANCH  |
| 69 | [73,211]   | [70,213]   | 2.414  | TIP     | BRANCH  |
| 70 | [346,213]  | [386,226]  | 55.899 | BRANCH  | BRANCH  |
| 71 | [331,215]  | [346,213]  | 15.657 | TIP     | BRANCH  |
| 72 | [346,213]  | [386,226]  | 48.728 | BRANCH  | BRANCH  |
| 73 | [14,169]   | [24,214]   | 48.142 | TIP     | BRANCH  |
| 74 | [24,214]   | [15,219]   | 9.657  | BRANCH  | BRANCH  |
| 75 | [1,206]    | [15,219]   | 21.899 | TIP     | BRANCH  |
| 76 | [15,219]   | [21,271]   | 54.314 | BRANCH  | BRANCH  |
| 77 | [386,226]  | [423,227]  | 36.828 | BRANCH  | BRANCH  |
| 78 | [423,227]  | [436,240]  | 18.728 | BRANCH  | BRANCH  |
| 79 | [491,228]  | [485,229]  | 5.414  | BRANCH  | BRANCH  |
| 80 | [491,228]  | [502,271]  | 46.142 | BRANCH  | BRANCH  |
| 81 | [464,223]  | [485,229]  | 25.556 | TIP     | BRANCH  |
| 82 | [485,229]  | [461,260]  | 39.527 | BRANCH  | BRANCH  |
| 83 | [447,243]  | [436,240]  | 11.243 | TIP     | BRANCH  |
| 84 | [436,240]  | [440,261]  | 21.657 | BRANCH  | BRANCH  |
| 85 | [63,240]   | [64,243]   | 2.414  | TIP     | BRANCH  |
| 86 | [64,243]   | [79,259]   | 25.899 | BRANCH  | BRANCH  |
| 87 | [86,262]   | [79,259]   | 7.243  | TIP     | BRANCH  |
| 88 | [69,264]   | [79,259]   | 11.071 | TIP     | BRANCH  |
| 89 | [461,260]  | [440,261]  | 20.414 | BRANCH  | BRANCH  |
| 90 | [461,260]  | [475,285]  | 29.385 | BRANCH  | BRANCH  |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 91  | [440,261]  | [436,278]  | 17.243 | BRANCH  | BRANCH  |
| 92  | [2,261]    | [8,271]    | 11.485 | TIP     | BRANCH  |
| 93  | [8,271]    | [21,271]   | 12.0   | BRANCH  | BRANCH  |
| 94  | [2,279]    | [8,271]    | 9.071  | TIP     | BRANCH  |
| 95  | [21,271]   | [25,277]   | 6.243  | BRANCH  | BRANCH  |
| 96  | [502,271]  | [483,285]  | 24.385 | BRANCH  | BRANCH  |
| 97  | [502,271]  | [503,296]  | 27.314 | BRANCH  | BRANCH  |
| 98  | [52,296]   | [25,277]   | 38.556 | TIP     | BRANCH  |
| 99  | [2,326]    | [25,277]   | 57.527 | TIP     | BRANCH  |
| 100 | [426,278]  | [436,278]  | 9.0    | BRANCH  | BRANCH  |
| 101 | [426,278]  | [396,324]  | 60.355 | BRANCH  | BRANCH  |
| 102 | [426,278]  | [370,281]  | 92.841 | BRANCH  | BRANCH  |
| 103 | [443,283]  | [436,278]  | 8.657  | TIP     | BRANCH  |
| 104 | [351,280]  | [370,281]  | 19.243 | TIP     | BRANCH  |
| 105 | [370,281]  | [369,296]  | 15.243 | BRANCH  | BRANCH  |
| 106 | [475,285]  | [483,285]  | 7.0    | BRANCH  | BRANCH  |
| 107 | [475,285]  | [436,322]  | 58.77  | BRANCH  | BRANCH  |
| 108 | [483,285]  | [503,296]  | 24.314 | BRANCH  | BRANCH  |
| 109 | [329,310]  | [369,296]  | 44.799 | TIP     | BRANCH  |
| 110 | [369,296]  | [394,324]  | 41.042 | BRANCH  | BRANCH  |
| 111 | [505,301]  | [503,296]  | 4.828  | TIP     | BRANCH  |
| 112 | [432,323]  | [436,322]  | 3.414  | TIP     | BRANCH  |
| 113 | [439,324]  | [436,322]  | 2.414  | TIP     | BRANCH  |
| 114 | [394,324]  | [396,324]  | 1.0    | BRANCH  | BRANCH  |
| 115 | [394,324]  | [381,354]  | 33.971 | BRANCH  | BRANCH  |
| 116 | [396,324]  | [422,345]  | 34.456 | BRANCH  | BRANCH  |
| 117 | [82,299]   | [82,331]   | 34.314 | TIP     | BRANCH  |
| 118 | [98,329]   | [82,331]   | 15.828 | TIP     | BRANCH  |
| 119 | [82,331]   | [57,351]   | 32.456 | BRANCH  | BRANCH  |
| 120 | [470,336]  | [469,340]  | 3.414  | TIP     | BRANCH  |
| 121 | [469,340]  | [495,352]  | 29.971 | BRANCH  | BRANCH  |
| 122 | [469,340]  | [436,341]  | 36.314 | BRANCH  | BRANCH  |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 123 | [436,341]  | [422,345]  | 14.657 | BRANCH  | BRANCH  |
| 124 | [436,341]  | [435,354]  | 12.828 | BRANCH  | BRANCH  |
| 125 | [422,345]  | [435,354]  | 17.485 | BRANCH  | BRANCH  |
| 126 | [57,351]   | [5,357]    | 56.799 | BRANCH  | BRANCH  |
| 127 | [57,351]   | [40,385]   | 40.87  | BRANCH  | BRANCH  |
| 128 | [495,352]  | [500,352]  | 4.0    | BRANCH  | BRANCH  |
| 129 | [495,352]  | [486,380]  | 30.314 | BRANCH  | BRANCH  |
| 130 | [490,319]  | [500,352]  | 38.627 | TIP     | BRANCH  |
| 131 | [509,362]  | [500,352]  | 12.314 | TIP     | BRANCH  |
| 132 | [398,354]  | [381,354]  | 16.828 | TIP     | BRANCH  |
| 133 | [381,354]  | [388,380]  | 38.385 | BRANCH  | BRANCH  |
| 134 | [436,356]  | [435,354]  | 1.414  | TIP     | BRANCH  |
| 135 | [5,357]    | [6,382]    | 26.899 | BRANCH  | BRANCH  |
| 136 | [3,354]    | [5,357]    | 2.414  | TIP     | BRANCH  |
| 137 | [373,402]  | [388,380]  | 27.799 | TIP     | BRANCH  |
| 138 | [388,380]  | [414,400]  | 32.87  | BRANCH  | BRANCH  |
| 139 | [486,380]  | [456,382]  | 29.828 | BRANCH  | BRANCH  |
| 140 | [509,403]  | [486,380]  | 35.213 | TIP     | BRANCH  |
| 141 | [6,382]    | [40,385]   | 35.485 | BRANCH  | BRANCH  |
| 142 | [1,391]    | [6,382]    | 9.657  | TIP     | BRANCH  |
| 143 | [450,372]  | [456,382]  | 11.485 | TIP     | BRANCH  |
| 144 | [456,382]  | [428,419]  | 49.355 | BRANCH  | BRANCH  |
| 145 | [43,395]   | [40,385]   | 9.828  | TIP     | BRANCH  |
| 146 | [414,400]  | [428,419]  | 27.314 | BRANCH  | BRANCH  |
| 147 | [426,376]  | [414,400]  | 27.556 | TIP     | BRANCH  |
| 148 | [362,338]  | [327,408]  | 85.74  | TIP     | BRANCH  |
| 149 | [360,396]  | [327,408]  | 37.799 | TIP     | BRANCH  |
| 150 | [320,435]  | [327,408]  | 30.556 | TIP     | BRANCH  |
| 151 | [35,406]   | [29,417]   | 13.657 | TIP     | BRANCH  |
| 152 | [24,399]   | [29,417]   | 21.899 | TIP     | BRANCH  |
| 153 | [24,443]   | [29,417]   | 27.071 | TIP     | BRANCH  |
| 154 | [428,419]  | [428,443]  | 23.0   | BRANCH  | BRANCH  |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 155 | [369,408]  | [364,429]  | 22.071 | TIP     | BRANCH  |
| 156 | [364,429]  | [383,429]  | 18.0   | BRANCH  | BRANCH  |
| 157 | [331,451]  | [364,429]  | 42.87  | TIP     | BRANCH  |
| 158 | [405,420]  | [383,429]  | 24.728 | TIP     | BRANCH  |
| 159 | [383,429]  | [383,445]  | 15.0   | BRANCH  | BRANCH  |
| 160 | [298,434]  | [285,438]  | 13.657 | TIP     | BRANCH  |
| 161 | [266,437]  | [285,438]  | 19.243 | TIP     | BRANCH  |
| 162 | [281,454]  | [285,438]  | 16.657 | TIP     | BRANCH  |
| 163 | [411,448]  | [428,443]  | 18.071 | TIP     | BRANCH  |
| 164 | [428,443]  | [468,484]  | 60.255 | BRANCH  | BRANCH  |
| 165 | [367,464]  | [383,445]  | 26.385 | TIP     | BRANCH  |
| 166 | [383,445]  | [389,448]  | 5.828  | BRANCH  | BRANCH  |
| 167 | [403,447]  | [389,448]  | 14.243 | TIP     | BRANCH  |
| 168 | [389,448]  | [393,472]  | 24.657 | BRANCH  | BRANCH  |
| 169 | [403,472]  | [393,472]  | 9.828  | TIP     | BRANCH  |
| 170 | [393,472]  | [382,480]  | 14.485 | BRANCH  | BRANCH  |
| 171 | [333,489]  | [382,480]  | 68.598 | TIP     | BRANCH  |
| 172 | [380,493]  | [382,480]  | 12.828 | TIP     | BRANCH  |
| 173 | [468,484]  | [483,500]  | 25.314 | BRANCH  | BRANCH  |
| 174 | [452,455]  | [468,484]  | 89.669 | TIP     | BRANCH  |
| 175 | [483,503]  | [483,500]  | 2.0    | TIP     | BRANCH  |
| 176 | [510,481]  | [483,500]  | 36.385 | TIP     | BRANCH  |
| 177 | [118,0]    | [106,16]   | 22.728 | TIP     | TIP     |
| 178 | [239,0]    | [254,32]   | 39.627 | TIP     | TIP     |
| 179 | [362,0]    | [354,10]   | 13.899 | TIP     | TIP     |
| 180 | [392,35]   | [387,36]   | 5.414  | TIP     | TIP     |
| 181 | [238,42]   | [221,54]   | 22.556 | TIP     | TIP     |
| 182 | [380,46]   | [369,61]   | 19.556 | TIP     | TIP     |
| 183 | [189,89]   | [185,103]  | 15.657 | TIP     | TIP     |
| 184 | [217,106]  | [206,121]  | 19.556 | TIP     | TIP     |
| 185 | [182,122]  | [214,176]  | 81.598 | TIP     | TIP     |
| 186 | [310,131]  | [322,133]  | 12.828 | TIP     | TIP     |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 187 | [350,144]  | [361,183]  | 49.799 | TIP     | TIP     |
| 188 | [288,166]  | [294,190]  | 27.899 | TIP     | TIP     |
| 189 | [442,170]  | [428,177]  | 16.899 | TIP     | TIP     |
| 190 | [63,180]   | [64,189]   | 10.243 | TIP     | TIP     |
| 191 | [374,187]  | [412,203]  | 44.627 | TIP     | TIP     |
| 192 | [234,203]  | [234,204]  | 1.0    | TIP     | TIP     |
| 193 | [89,209]   | [93,213]   | 6.243  | TIP     | TIP     |
| 194 | [292,212]  | [268,232]  | 34.627 | TIP     | TIP     |
| 195 | [184,217]  | [186,219]  | 2.828  | TIP     | TIP     |
| 196 | [152,238]  | [171,239]  | 21.071 | TIP     | TIP     |
| 197 | [140,240]  | [121,242]  | 20.657 | TIP     | TIP     |
| 198 | [196,241]  | [188,257]  | 20.485 | TIP     | TIP     |
| 199 | [325,261]  | [295,282]  | 42.213 | TIP     | TIP     |
| 200 | [193,274]  | [212,298]  | 32.456 | TIP     | TIP     |
| 201 | [128,276]  | [146,276]  | 18.0   | TIP     | TIP     |
| 202 | [248,282]  | [267,295]  | 24.385 | TIP     | TIP     |
| 203 | [322,313]  | [292,324]  | 35.142 | TIP     | TIP     |
| 204 | [191,317]  | [186,343]  | 50.213 | TIP     | TIP     |
| 205 | [222,351]  | [228,374]  | 25.485 | TIP     | TIP     |
| 206 | [149,371]  | [151,376]  | 5.828  | TIP     | TIP     |
| 207 | [312,383]  | [291,392]  | 24.728 | TIP     | TIP     |
| 208 | [105,384]  | [83,395]   | 27.142 | TIP     | TIP     |
| 209 | [225,400]  | [222,411]  | 12.243 | TIP     | TIP     |
| 210 | [176,402]  | [183,416]  | 16.899 | TIP     | TIP     |
| 211 | [508,412]  | [473,426]  | 40.799 | TIP     | TIP     |
| 212 | [166,418]  | [134,432]  | 37.799 | TIP     | TIP     |
| 213 | [12,451]   | [20,509]   | 70.627 | TIP     | TIP     |
| 214 | [99,461]   | [100,486]  | 36.971 | TIP     | TIP     |
| 215 | [49,490]   | [40,508]   | 22.314 | TIP     | TIP     |
| 216 | [159,505]  | [154,510]  | 8.243  | TIP     | TIP     |
| 217 | [64,509]   | [73,510]   | 10.243 | TIP     | TIP     |